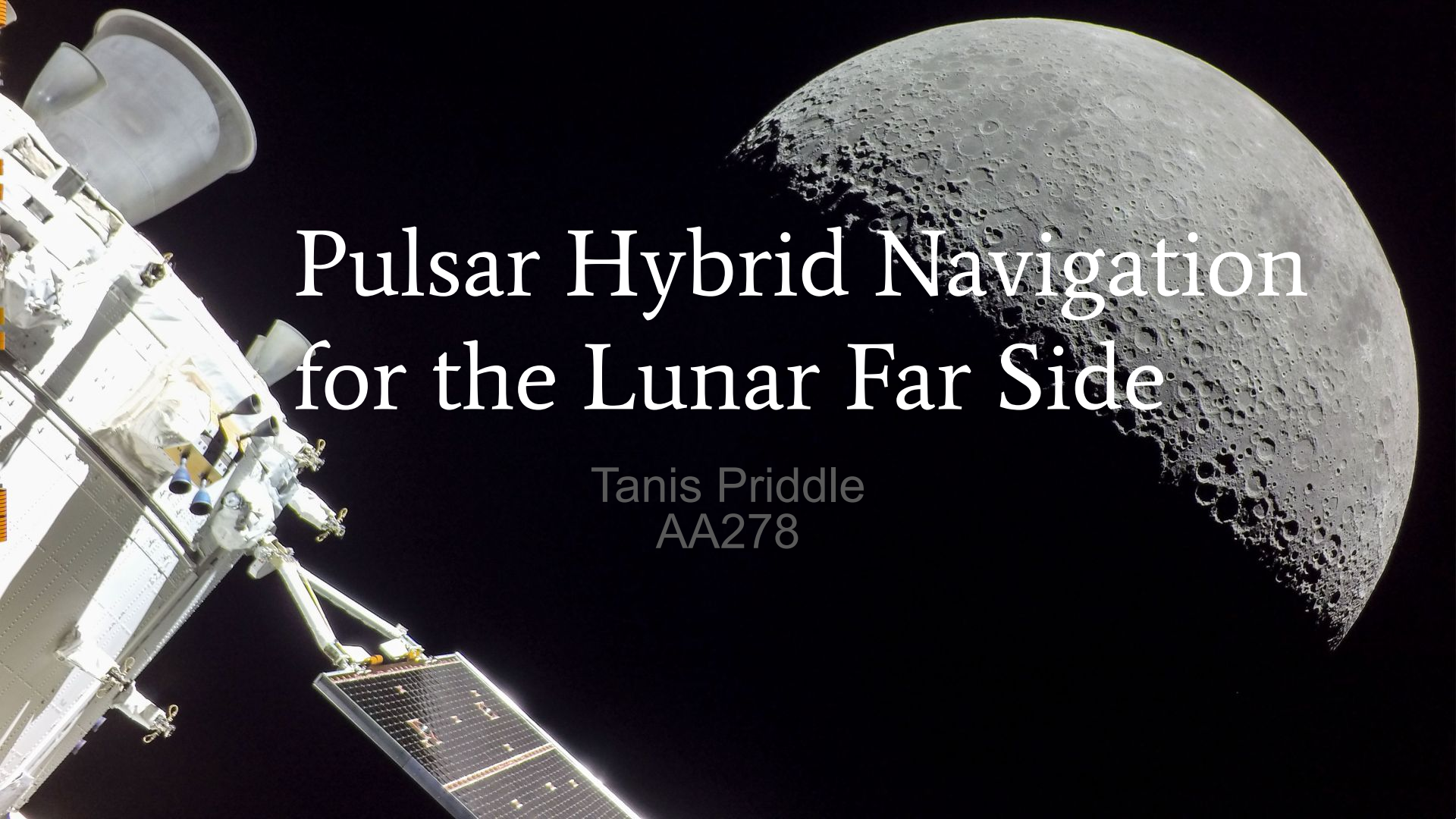


A satellite is shown in the foreground on the left, with its solar panels and various instruments visible. The background is a large, cratered Moon against a black sky.

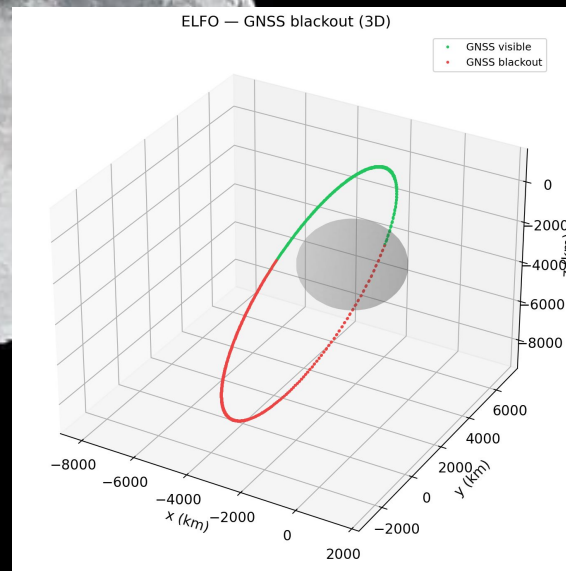
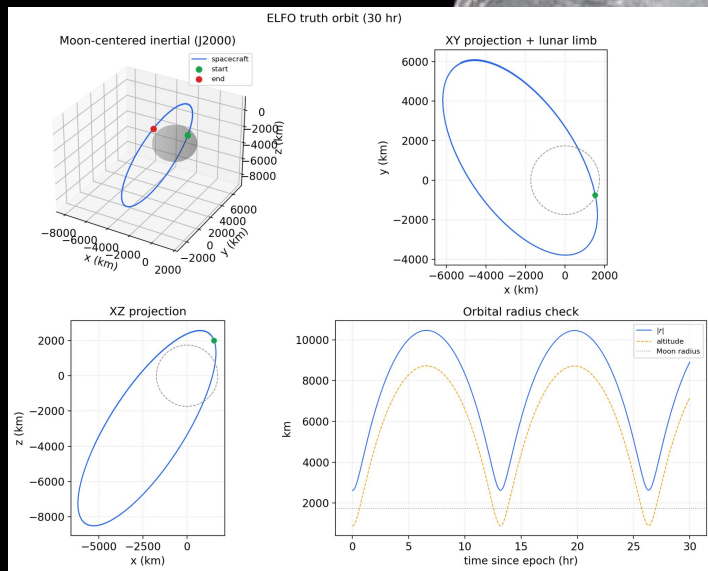
Pulsar Hybrid Navigation for the Lunar Far Side

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Design and evaluate a hybrid EKF that continuously fuses x-ray pulsar time-of-arrival measurements with LunaNet pseudorange fixes when available to maintain bounded position and timing accuracy for a lunar orbiter through far-side blackout periods where no Earth GNSS sidelobe signals are available and near-side navigation under poor sidelobe geometry.

- The lunar far side is permanently shielded from Earth -> no GNSS sidelobe signals reach it, leaving far-side orbiters with no absolute PNT source during blackouts that can last hours per orbit
- Far-side missions are a reality and scientifically critical, but the current ODTS achieves only 1.5 km accuracy even on the near side
- X-ray pulsars are a natural absolute positioning source available anywhere in deep space, are permanently visible regardless of orbital phase, and require no additional lunar infrastructure!

Can fusing X-ray pulsars, GNSS sidelobe signals, and LunaNet relays maintain continuous PNT through far-side GNSS blackout?



Approach

30-hour
ELFO
orbit
simulator
for truth
trajectory

Visibility
window
modeling
using
Earth-elevati
on masks

Define and
take
measurements
(XNAV, GNSS
sidelobe,
LunaNet
pseudorange)

10-state
EKF
filter

Monte Carlo
analysis over
3 policies:
XNAV-only,
GNSS-only,
hybrid

The Math

Algorithm 1 Hybrid Lunar Far-Side Navigation Pipeline

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1: Input: ELFO/LLO seed orbit; pulsar catalog  $\{\hat{\mathbf{n}}_k, \sigma_{\text{TOA},k}\}$ ; LunaNet Walker config; noise  $(\sigma_{\text{TOA}}, \sigma_{\text{GNSS}}, \sigma_{\text{LN}})$ ; policy  $\pi \in \{\text{XNAV}, \text{GNSS}, \text{HYBRID}\}$ ; trials  $M$ ; RNG
2: Output: Per-epoch  $\hat{\mathbf{x}}_i, P_i$ , position error  $e_i$ , NIS $_i$ , segment labels; Monte Carlo summary (final  $\mu/\text{p95}$ , RMS, segment means)

3: Truth propagation (MCI, J2000)
4: Integrate Moon-centered force (lunar gravity, Earth/Sun third-body,  $J_2$ , SRP) with SPICE DE440
5: for each epoch  $i = 0, \dots, N$  at step  $\Delta t$  do
6:   Store  $(t_i, \mathbf{r}_i^{\text{tru}}, \mathbf{v}_i^{\text{tru}}, \text{eti})$ ; map MCI  $\leftrightarrow$  ICRS
7: end for

8: Visibility and blackout labeling
9: for each epoch  $i$  do
10:   Earth elevation  $\varepsilon_i$ ; Moon-occultation test  $\text{occ}_i$ 
11:    $\text{GNSSVis}_i \leftarrow (\varepsilon_i \geq 5^\circ) \wedge \neg \text{occ}_i$ 
12:   Propagate LunaNet relays; get  $\text{LNVis}_i, n_{\text{LN},i}$ 
13:    $\text{BLACKOUT}_i \leftarrow \neg \text{GNSSVis}_i$ ; assign  $\text{NAVMode}_i$ 
14: end for
15: Stitch contiguous  $\text{BLACKOUT}_i$  into windows

16: for each trial  $m = 1, \dots, M$  do
17:   Filter init: draw  $\delta_0, \|\delta\| \sim \mathcal{U}[30, 100]$  km;  $\hat{\mathbf{r}}_0 \leftarrow \mathbf{r}_0^{\text{tru}} + \delta, \hat{\mathbf{v}}_0 \leftarrow \mathbf{v}_0^{\text{tru}}$ 
18:    $\hat{\mathbf{x}}_0 \leftarrow [\hat{\mathbf{r}}_0^\top, \hat{\mathbf{v}}_0^\top, 0, 0, 0, 0]^\top$ ;  $P_0 \leftarrow \text{diag}(\sigma_r^2 \mathbf{I}_3, \sigma_v^2 \mathbf{I}_3, \sigma_b^2, \sigma_b^2, \sigma_g^2, \sigma_g^2)$ 

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19: for each epoch  $i = 1, \dots, N$  do
20:   Predict:  $\hat{\mathbf{x}}_i \leftarrow \Phi \hat{\mathbf{x}}_{i-1}, P_i \leftarrow \Phi P_{i-1} \Phi^\top + Q$  (CV; CWNA  $Q$ ; oracle variant uses  $\mathbf{v}_{i-1}^{\text{tru}}$ )
21:   Schedule measurements  $\mathcal{M}_i$  by policy:
22:   if  $\pi = \text{XNAV}$  or  $\text{BLACKOUT}_i$  then
23:     Add pulsars:  $z_k = \hat{\mathbf{n}}_k \cdot \mathbf{r}_i^{\text{tru}} + \nu_k, \sigma_{z,k} = c\sigma_{\text{TOA},k}$ 
24:     If  $\text{HYBRID}$  and  $\text{LNVis}_i$ : add LunaNet pseudoranges
25:   else
26:     Sidelobe gate: keep PRN  $p$  iff  $\neg \text{occ}_E(p) \wedge \|\mathbf{r}_p - \mathbf{r}_{sc}\| > \|\mathbf{r}_E - \mathbf{r}_{sc}\| \wedge \ell_p \leq 6^\circ$ 
27:     Add GNSS pseudoranges (light-time):  $\rho = \|\mathbf{r}_{sc} - \mathbf{r}_{tx}(t_{tx})\| + b_{rx} - b_{tx} + \eta$ 
28:     if  $\pi = \text{HYBRID}$  then
29:       Also add LunaNet (if  $\text{LNVis}_i$ ) and pulsars
30:     else if  $\pi = \text{GNSS}$  and  $|\text{PRN}| = 0$  then
31:       Fall back to pulsars
32:     end if
33:   end if
34:   Joint update: stack  $H, R$  over  $\mathcal{M}_i$ ;  $S = H P_i H^\top + R$ ;  $K = P_i H^\top S^{-1}$ ;  $\hat{\mathbf{x}}_i \leftarrow K \mathbf{y}$ ;  $P_i \leftarrow (I - KH) P_i$ 
35:    $\text{NIS}_i \leftarrow \mathbf{y}^\top S^{-1} \mathbf{y}$ ;  $e_i \leftarrow \|\hat{\mathbf{r}}_i - \mathbf{r}_i^{\text{tru}}\|$ 
36: end for
37: end for

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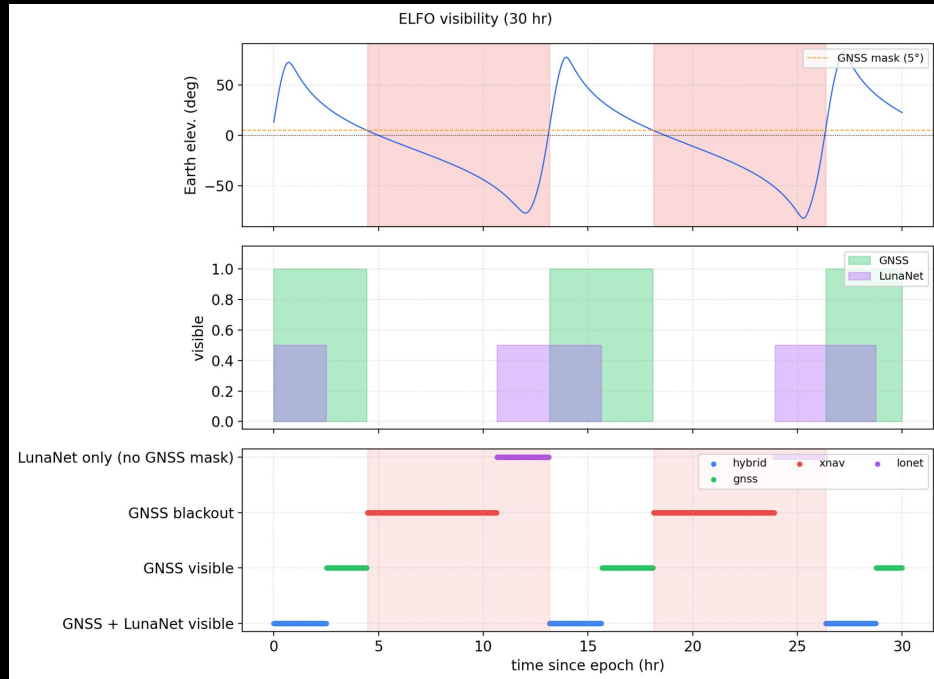
38: Monte Carlo aggregation: final mean/p95, arc RMS =  $\sqrt{\frac{1}{N} \sum_i e_i^2}$ 
39: Segment means  $\mu_{\text{blk}}, \mu_{\overline{\text{blk}}}$  vs. LunaNet ref (13.43 m); median NIS $_i/\text{dof}$ ; sweeps over pulsar count and  $\sigma_{\text{TOA}}$ 
40: Export per-trial CSVs, summary tables, figures

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Disclaimer! Assumptions....

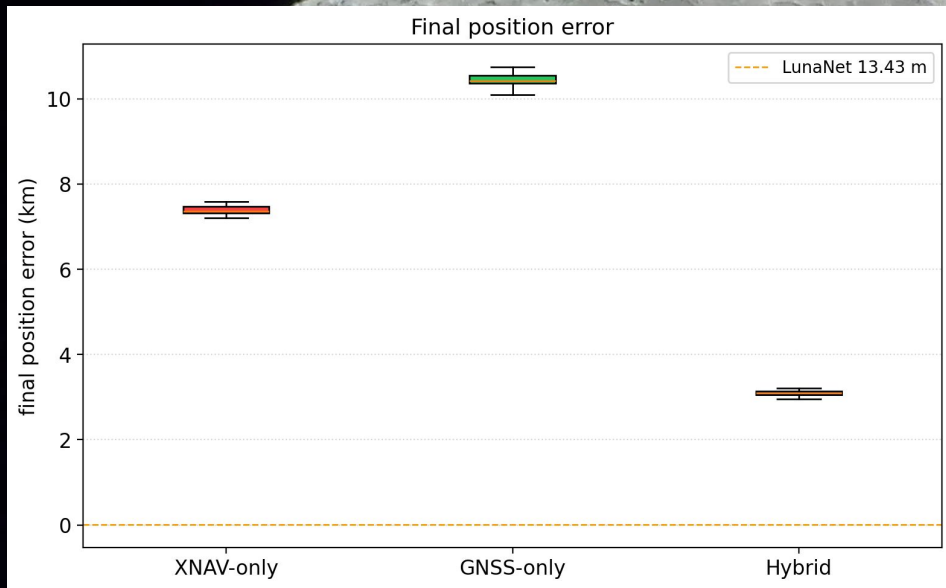
- Pulsar directions are treated as exactly known. Integer cycle phase ambiguity and timing-model error are not modeled.
- Measurements are synthesized from the truth trajectory plus noise, so there is no systematic bias -> absolute errors are OPTIMISTIC
- Truth and filter share the same measurement model
- Filter uses a constant-velocity dynamics model with continuous white-noise-acceleration process term -> unmodeled lunar gravity is absorbed by noise
- Sidelobe GNSS availability is reduced to an Earth-elevation/occultation test
- Transmitter clocks are treated as known in simulation, so estimated receiver clock bias is referenced to a perfect time scale
- Clock bias is observable only through pseudoranges
- Transmitter and relay ephemerides are known -> pseudorange error is due to measurement noise, not orbit error

Visibility and Policy Switching



- For this ELFO, Earth drops below the 5° mask or behind the lunar limb for **~64%** of the period
- Sidelobe GNSS is sparse even when Earth is visible! I consider only satellites past the Earth limb on the far side reach the Moon, giving **2-6 usable PRNs** with poor geometry.
- LunaNet relays fill some gaps but are themselves intermittent (Walker constellation visibility depends on orbital phase)
- REMINDER! The visibility *label* (what's geometrically available) is distinct from the filter policy (hybrid keeps pulsars in the update outside blackout to stabilize the weak sidelobe geometry)

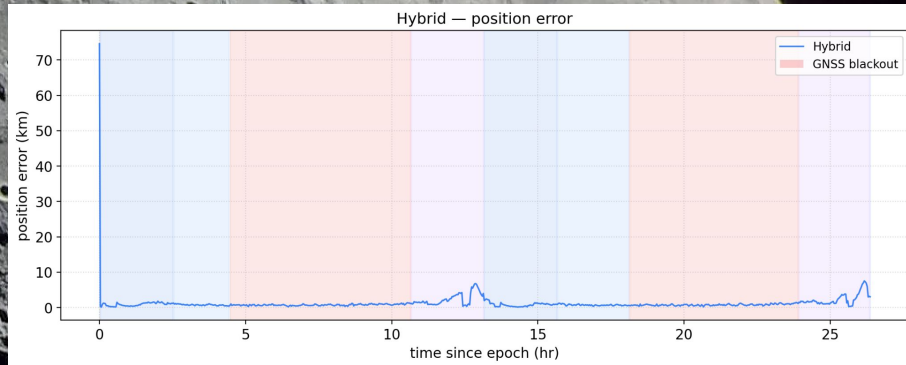
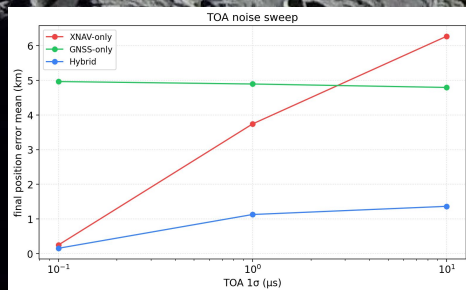
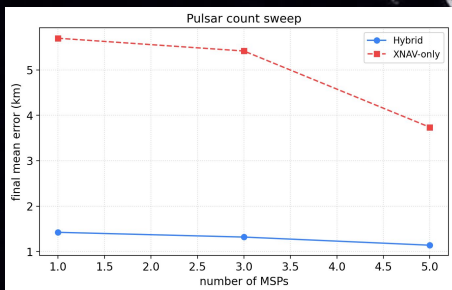
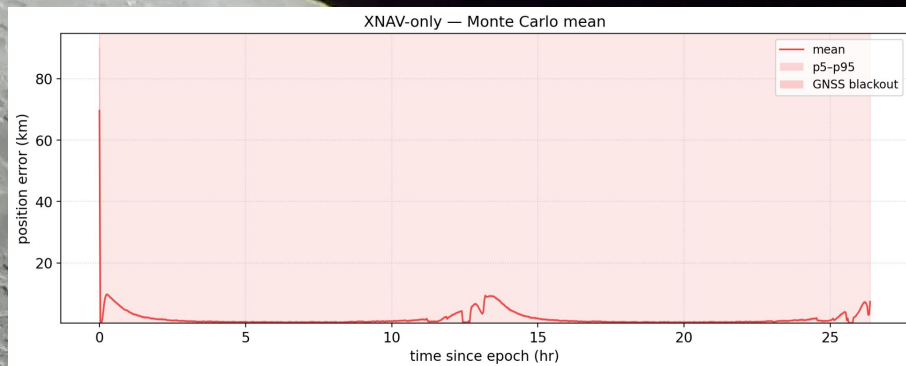
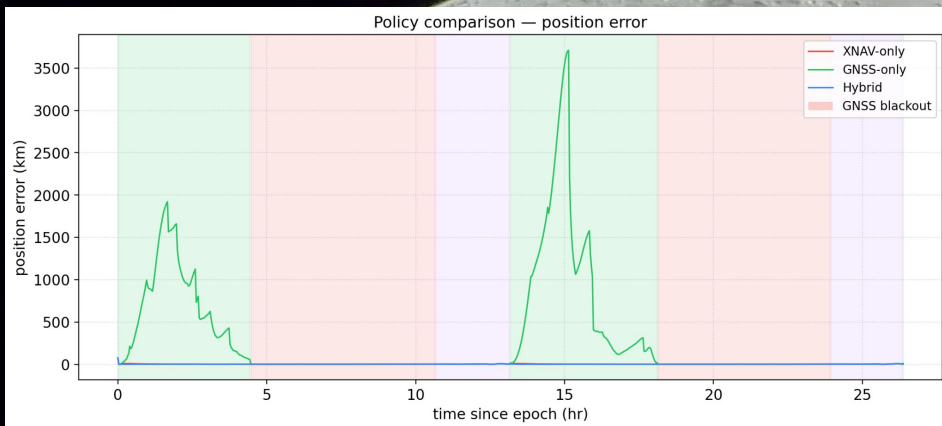
Results: Policy Comparison



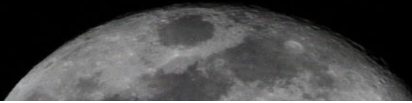
Policy	Final μ (km)	Final p95 (km)	RMS (km)	Blackout μ (km)	<13.43 m p95?
xnav_only	7.39	7.59	3.50	1.24	no
gnss_only	10.44	10.64	680.18	1.23	no
hybrid	3.09	3.17	2.78	1.23	no

- **Hybrid wins on every metric!**
- GNSS-only diverges
- XNAV-only stays bounded -> reliable fallback!
- Blackout μ is identical as expected: every policy falls back to the same pulsars

Realism, Sensitivity, and Limitations

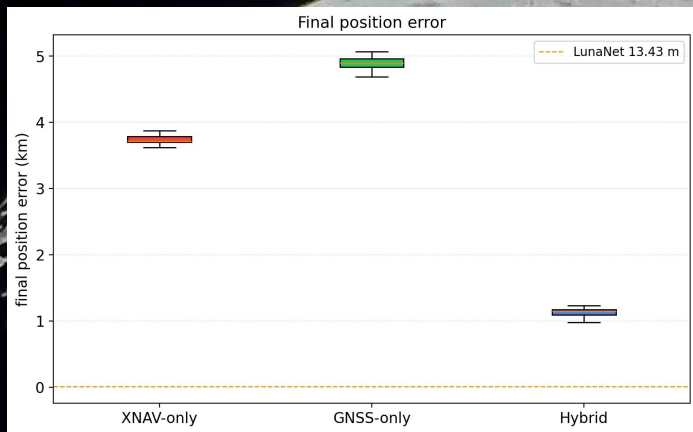


Effect of Process Noise

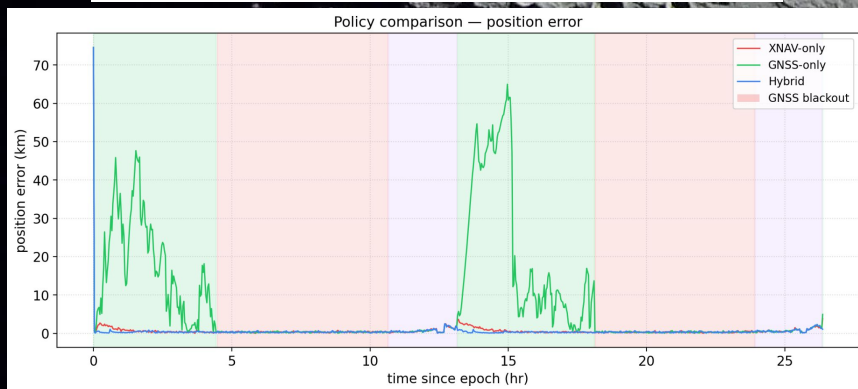


Policy	Q Model	Q Param	Final Mean	Final P95	RMS	Blackout Mean	Non-Blackout Mean	NIS/DOF
Hybrid	Constant	10^{-5}	13.135	13.167	6.410	4.678	3.668	75.15
GNSS Only	Constant	10^{-5}	26.640	26.675	1491.413	4.692	1977.798	133.54
Hybrid	Constant	10^{-4}	3.050	3.073	3.273	1.234	1.110	8.75
GNSS Only	Constant	10^{-4}	10.408	10.504	679.079	1.235	808.791	12.46
Hybrid	Constant	10^{-3}	0.553	0.573	2.951	0.429	0.553	1.69
GNSS Only	Constant	10^{-3}	4.936	5.071	260.592	0.431	285.484	2.01
Hybrid	Gravity-Scaled	0.5	0.043	0.076	2.938	0.347	0.510	1.20
GNSS Only	Gravity-Scaled	0.5	1.374	2.553	53.407	0.345	65.512	1.34
Hybrid	Gravity-Scaled	1.0	0.051	0.083	2.937	0.335	0.502	0.71
GNSS Only	Gravity-Scaled	1.0	1.429	3.111	24.945	0.334	28.992	0.68
Hybrid	Gravity-Scaled	2.0	0.053	0.085	2.938	0.342	0.505	0.55
GNSS Only	Gravity-Scaled	2.0	1.472	3.272	16.586	0.344	20.436	0.47

Aside: Truth Orbit Propagation



Policy	Final μ (km)	Final p95 (km)	RMS (km)	Blackout μ (km)	<13.43 m p95?
xnav_only	3.74	3.86	2.46	0.46	no
gnss_only	4.89	5.02	16.09	0.45	no
hybrid	1.13	1.23	2.39	0.45	no



- **Truth-velocity** filter propagates using the *true* velocity each step (removes dynamics error, isolates measurement quality)
Optimistic!
- **Filter CV** propagates on its own estimated velocity with no gravity
 - a. Weakest at perilune, where gravity is stronger
- *Relative ranking holds in both modes*

Conclusions and Future Work

- Hybrid fusion maintains far-side PNT when GNSS-only fails!
 - Pulsars are a reliable fallback, and LunaNet sharpens fixes when relays are up
 - Hybrid can be more accurate than GNSS-relay in Earth visibility
-
- ★ Add a per-pulsar systematic floor + realistic σ for flight-representative absolute error
 - ★ Gravity-scaled process noise for filter consistency
 - ★ Higher fidelity lunar gravity to reduce GNSS divergence
 - ★ Model pulsar phase/cycle ambiguity
 - ★ CV -> Full EKF acceleration integration...in the works!